ExamGOAL
PYQ PDF

Free · With answer key

GATE ECE · CONTROL SYSTEMS · CHAPTER-WISE PYQS

Stability

GATE ECE Control Systems · Previous Year Questions with Answer Key

42	31	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	14 questions
Marks 2	24 questions
Marks 5	4 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

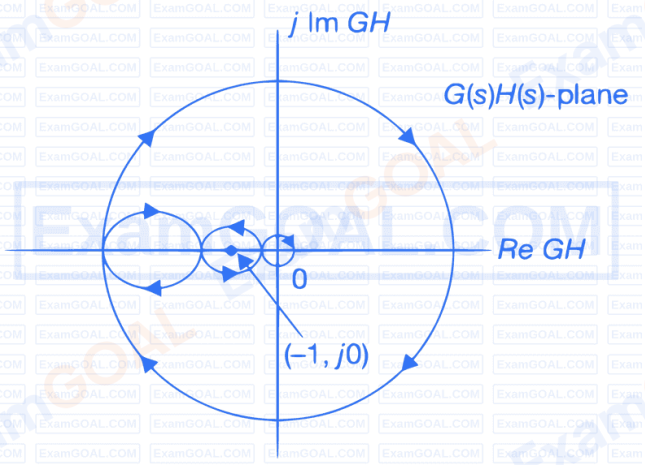
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 14 questions

Q1. GATE ECE 2021

Open solution

The complete Nyquist plot of the open-loop transfer function $G(s)H(s)$ of a feedback control system is shown in the figure.



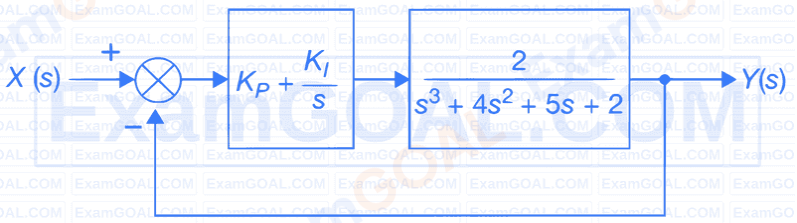
If $G(s)H(s)$ has one zero in the right-half of the s-plane, the number of poles that the closed-loop system will have in the right-half of the s-plane is

(A) 1 (B) 0 (C) 4 (D) 3

Q2. GATE ECE 2021

Open solution

A unity feedback system that uses proportional - integral (PI) control is shown in the figure.



The stability of the overall system is controlled by tuning the PI control parameters K_p and K_I The maximum value of K_I that can be chosen so as to keep the overall system stable or, in the worst case, marginally - stable (rounded off to three decimal places) is ___

Q3. GATE ECE 2020

Open solution

The loop transfer function of a negative feedback system is

$G(s)H(s) = \frac{K(s+11)}{s(s+2)(s+8)}$

The value of K, for which system is marginally stable, is ____ .

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q4. GATE ECE 2018

Consider p(s) = $s^3 + a_2 s^2 + a_1 s + a_0$ with all real coefficients. It is known that its derivative p'(s) has no real roots. The number of real roots of p(s) is

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q5. GATE ECE 2018

The Nyquist stability criterion and the Routh criterion both are powerful analysis tools for determining the stability of feedback controllers. Identify which of the following statements is FALSE.

(A) The closed-loop frequency response for a unity feedback system cannot be obtained from the Nyquist plot.

(B) The Routh criterion is not applicable in the condition of transport lag, which can be readily handled by the Nyquist criterion.

(C) The general shape of the Nyquist plot is readily obtained from the Bode magnitude plot for all minimum-phase systems.

(D) Both the criteria provide information relative to the stable gain range of the system.

Open solution

Q6. GATE ECE 2016 Set 1

Match the inferences X, Y, and Z, about a system, to the corresponding properties of the elements of first column in Routh's Table of the system characteristic equation.

X: The system is stable ...
Y: The system is unstable ...
Z: The test breaks down ...

P: ... when all elements are positive
Q: ... when any one element is zero
R: ... when there is a change in sign of coefficients

(A) X → P, Y → Q, Z → R (B) X → Q, Y → P, Z → R (C) X → R, Y → Q, Z → P (D) X → P, Y → R, Z → Q

Open solution

Q7. GATE ECE 2015 Set 1

A unity negative feedback system has the open-loop transfer function

$$G(s) = \frac{K}{s(s+1)(s+3)}$$

The value of the gain K (>0) at which the root locus crosses the imaginary axis is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q8. GATE ECE 2014 Set 1

The forward path transfer function of a unity negative feedback system is given by

$$G(s) = \frac{k}{(s+2)(s-1)}$$

The value of K which will place both the poles of the closed-loop system at the same location, is _____.

Open solution

Q9. GATE ECE 2013

Which one of the following statements is NOT TRUE for a continuous time causal and stable LTI system?

(A) All the poles of the system must lie on the left side of the jω axis

(B) Zeros of the system can lie anywhere in the s-plane

(C) All the poles must lie within $|s| = 1$

(D) All the roots of the characteristic equation must be located on the left side of the jω axis

Open solution

Q10. GATE ECE 2013

A polynomial $f(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x - a_0$ with all coefficients positive has

(A) no real roots

(B) no negative real root

(C) odd number of real roots

(D) at least one positive and one negative real root

Open solution

Q11. GATE ECE 2000

An amplifier with resistive negative feedback has two left half-plane poles in its open-loop transfer function. The amplifier

(A) will always be unstable at high frequencies

(B) will be stable for all frequencies

(C) may be unstable, depending on the feedback factor

(D) will oscillate at low frequencies

Open solution

Q12. GATE ECE 2000

A system described by the transfer function

$$H(s) = \frac{1}{s^3 + \alpha s^2 + ks + 3}$$

is stable. The constraints on α and k are,

(A) $\alpha > 0,\ \alpha k < 3$

(B) $\alpha > 0,\ \alpha k > 3$

(C) $\alpha < 0,\ \alpha k > 3$

(D) $\alpha > 0,\ \alpha k < 3$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. GATE ECE 1998

The number of roots of $s^3 + 5s^2 + 7s + 3 = 0$ in the left half of the s-plane are

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q14. GATE ECE 1998

The open loop transfer function of a unity feedback open-loop system is $\frac{2s^2+6s+5}{(s+1)^2(s+2)}$. The characteristic equation of the closed loop system is

(A) $2s^2 + 6s + 5 = 0$ (B) $(s+1)^2(s+2) = 0$

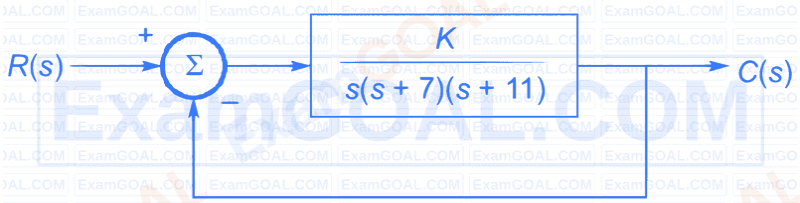
(C) $2s^2 + 6s + 5 + (s+1)^2(s+2) = 0$ (D) $2s^2 + 6s + 5 - (s+1)^2(s+2) = 0$

Open solution

Marks 2 24 questions

Q15. GATE ECE 2026

Consider the unity negative feedback control system shown in the Figure. The value of gain $K(>0)$ at which the given system will remain marginally stable is ____ . (Answer in integer)



Open solution

Q16. GATE ECE 2025

Consider the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. Let (L, I, R) be defined as follows. L is the number of roots of $p(s)$ with negative real parts. I is the number of roots of $p(s)$ that are purely imaginary. R is the number of roots of $p(s)$ with positive real parts. Which one of the following options is correct?

(A) $L = 2, I = 2$ and $R = 1$ (B) $L = 3, I = 2$ and $R = 0$

(C) $L = 1, I = 2$ and $R = 2$ (D) $L = 0, I = 4$ and $R = 1$

Open solution

Q17. GATE ECE 2022

Consider an even polynomial p(s) given by

$p(s) = s^4 + 5s^2 + 4 + K$

where K is an unknown real parameter. The complete range of K for which p(s) has all its roots on the imaginary axis is _____.

(A) $-4 \le K \le \frac{9}{4}$ (B) $-3 \le K \le \frac{9}{2}$ (C) $-6 \le K \le \frac{5}{4}$ (D) $-5 \le K \le 0$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q18. GATE ECE 2017 Set 2

A unity feedback control system is characterized by the open-loop transfer function

$$G(s) = \frac{2(s+1)}{s^3 + ks^2 + 2s + 1}$$

The value of k for which the system oscillates at 2 rad/s is _____.

Open solution

Q19. GATE ECE 2017 Set 1

Which one of the following options correctly describes the locations of the roots of the equation $s^4 + s^2 + 1 = 0$ on the complex plane?

(A) Four left half plane (LHP) roots

(B) One right half plane (RHP) root, one LHP root and two roots on the imaginary axis

(C) Two RHP roots and two LHP roots

(D) All four roots are on the imaginary axis

Open solution

Q20. GATE ECE 2016 Set 1

The transfer function of a linear time invariant system is given by $H(s) = 2s^4 - 5s^3 + 5s - 2$. The number of zeros in the right half of the s-plane is _____.

Open solution

Q21. GATE ECE 2016 Set 3

The first two rows in the Routh table for the characteristic equation of a certain closed-loop control system are given as

s^3	1	$(2K+3)$
s^2	$2K$	4

ExamSide.Com

The range of K for which the system is stable is

(A) -2.0 < K < 0.5 (B) 0 < K < 0.5 (C) 0 < K < ∞ (D) 0.5 < K < ∞

Open solution

Q22. GATE ECE 2015 Set 3

The characteristic equation of an LTI system is given by

$F(s) = s^5 + 2s^4 + 3s^3 + 6s^2 - 4s - 8 = 0$. The number of roots that lie strictly in the left half s-plane is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q23. GATE ECE 2015 Set 1

Open solution

A plant transfer function is given as

$$G(s) = \left(K_p + \frac{K_1}{s}\right)\left(\frac{1}{s(s+2)}\right)$$

. When the plant operates in a unity feedback configuration, the condition for the stability of the closed loop system is

(A) $K_p > \frac{K_1}{2} > 0$ (B) $2K_1 > K_p > 0$ (C) $2K_1 < K_p$ (D) $2K_1 > K_p$

Q24. GATE ECE 2014 Set 4

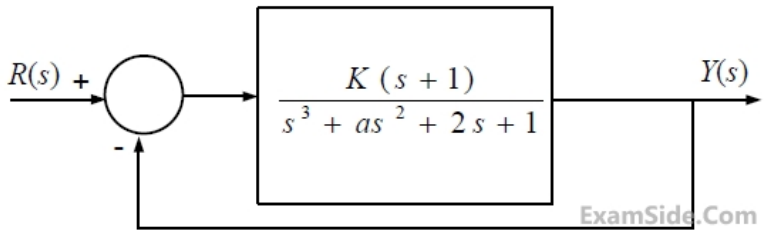
Open solution

Consider a transfer function $G_p(s) = \frac{ps^2+3ps-2}{s^2+(3+p)s+(2-p)}$ with 'p' a positive real parameter. The maximum value of 'p' until which G_p remains stable is _____.

Q25. GATE ECE 2012

Open solution

The feedback system shown below oscillates at 2 rad/s when



(A) K = 2 and a = 0.75 (B) K = 3 and a = 0.75
(C) K = 4 and a = 0.5 (D) K = 2 and a = 0.5

Q26. GATE ECE 2008

Open solution

The number of open right half plane poles of

$$G(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3}\ is$$

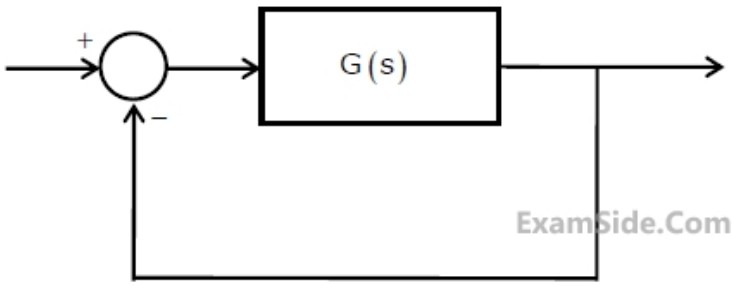
(A) 0 (B) 1 (C) 2 (D) 3

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q27. GATE ECE 2008

Open solution

A certain system has transfer function $G(s) = \frac{s+8}{s^2+\alpha s-4}$, where α is a parameter. Consider the standard negative unity feedback configuration as shown below



Which of the following statements is true?

(A) The closed loop system is never stable for any value of α

(B) For some positive values of α, the closed loop system is stable, but not for all positive values

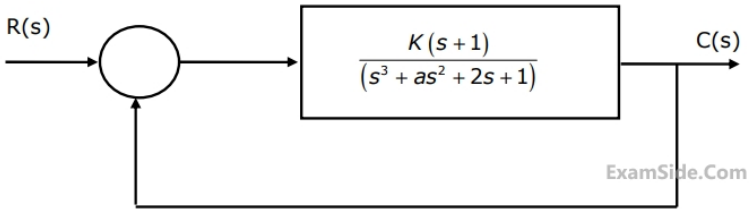
(C) For all positive values of α, the closed loop system is stable

(D) The closed loop system is stable for all values of α, both positive and negative

Q28. GATE ECE 2006

Open solution

The positive values of "K" and "a" so that the system shown in the figure below oscillates at a frequency of 2 rad/sec respectively are



(A) 1, 0.75 (B) 2, 0.75 (C) 1, 1 (D) 2, 2

Q29. GATE ECE 2004

Open solution

For the polynomial

$P(s) = s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15$,

the number of roots which lie in the right half of the s-plane is

(A) 4 (B) 2 (C) 3 (D) 1

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q30. GATE ECE 2003

The open-loop transfer function of a unity feedback system is

$$G(s) = \frac{k}{s(s^2 + s + 2)(s + 3)}$$

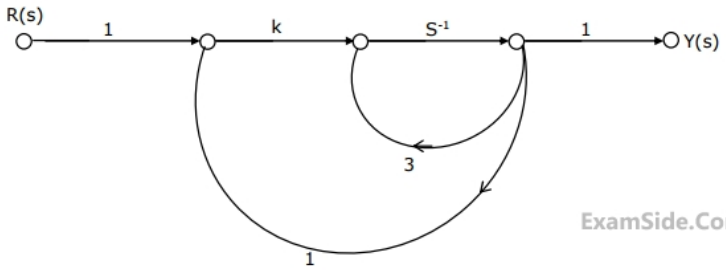
the range of 'k' for which the system is stable

(A) $\frac{21}{4} > k > 0$ (B) 13 > k > 0 (C) $\frac{21}{4} < k < \infty$ (D) $-6 < k < \infty$

Open solution

Q31. GATE ECE 2002

The system shown in Figure remains stable when



ExamSide.Com

(A) k<-1 (B) -1 < k < 1 (C) 1 < k < 3 (D) k > 3

Open solution

Q32. GATE ECE 2002

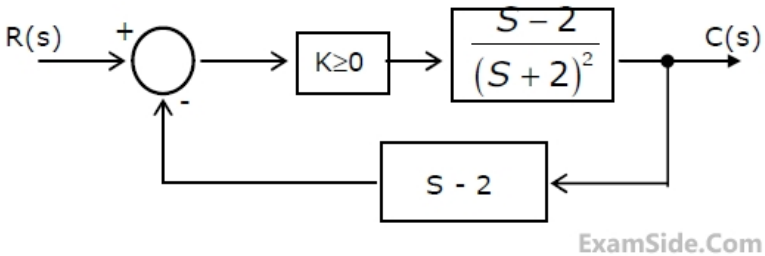
The characteristic polynomial of a system is
$q(s) = 2s^5 + s^4 + 4s^3 + 2s^2 + 2s + 1$. The system is

(A) stable (B) marginally stable
(C) unstable (D) oscillatory

Open solution

Q33. GATE ECE 2001

The feedback control system in Figure is stable



ExamSide.Com

(A) for all K $\geq$ 0 (B) only if K $\geq$ 1
(C) only if 0 $\leq$ K < 1 (D) only if 0 $\leq$ K $\leq$ 1

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q34. GATE ECE 1994

If G(s) is a stable transfer function, then $F(s) = \frac{1}{G(s)}$ is always a stable transfer function.

(A) TRUE (B) FALSE

Open solution

Q35. GATE ECE 1993

If $s^3 + 3s^2 + 4s + A = 0$,then all the roots of this equation are in the left half plane provided that

(A) A > 12 (B) -3 < A < 4 (C) 0 < A < 12 (D) 5 < A < 12

Open solution

Q36. GATE ECE 1990

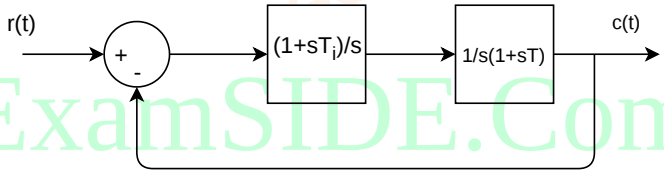
An electromechanical closed-loop control system has the following characteristic equation $s^3 + 6Ks^2 + (K+2)s + 8 = 0$, where K is the forward gain of the system.The condition for closed loop stability is

(A) K = 0.528 (B) K = 2 (C) K = 0 (D) K = -2.258

Open solution

Q37. GATE ECE 1989

In order to stabilize the system shown in fig. T_i should satisfy:



ExamSIDE.Com

(A) $T_i = -T$ (B) $T_i = T$.

(C) $T_i < T$. (D) $T_i > T$.

Open solution

Q38. GATE ECE 1988

Consider a characteristic equation given by

$s^4 + 3s^3 + 5s^2 + 6s + K + 10 = 0$.

The condition for stability is

(A) K > 5 (B) -10 < K (C) K > -4 (D) -10 < K < -4

Open solution

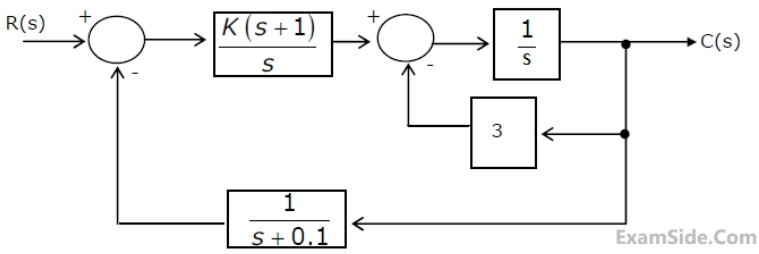
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Marks 5 4 questions

Q39. GATE ECE 2001

Open solution

Consider the feedback control system shown in figure.



(a) Find the transfer function of the system and its characteristic equation.
(b) Use the Routh-Hurwitz criterion to determine the range of 'K' for which the system is stable.

Q40. GATE ECE 1999

Open solution

The loop transfer function of a feedback control system is given by

$$G(s)H(s) = \frac{K(s+1)}{s(1+Ts)(1+2s)},\ K > 0$$

Using Routh-Hurwitz criterion, determine the region of K-T plane in which the closed-loop system is stable.

Q41. GATE ECE 1998

Open solution

The characteristic equation of a feedback control system is

$$s^4 + 20s^3 + 15s^2 + 2s + K = 0$$

(i) Determine the range of K for the system to be stable.
(ii) Can the system be marginally stable? If so, find the required value of K and the frequency of sustained oscillation.

Q42. GATE ECE 1996

Open solution

A system having an open loop transfer function $G(s) = \frac{K(s+3)}{s(s^2+2s+2)}$ is used in a control system with unity negative feedback. Using the Routh-Hurwitz criterion, find the range of values of 'K' for which the feedback system is stable.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Marks 1

1. **D**	2. **3.1 to 3.2**	3. **160**	4. **B**	5. **A**
6. **D**	7. **12**	8. **2.25**	9. **C**	10. **D**
11. **B**	12. **B**	13. **D**	14. **C**	

Marks 2

15. **(1386 to 1386)**	16. **A**	17. **A**	18. **0.75**	19. **C**
20. **3**	21. **D**	22. **2**	23. **A**	24. **2**
25. **A**	26. **C**	27. **C**	28. **B**	29. **B**
30. **A**	31. **D**	32. **C**	33. **C**	34. **B**
35. **C**	36. **B**	37. **D**	38. **D**	

Marks 5

39. **(a) Transfer function =** $\frac{K(s+1)(s+0.1)}{s^3+3.1s^2+s(K+0.3)+K}$ **Characteristic equation =** $s^3+3.1s^2+s\,(K+0.3)+K=0$ **(b) For K > 0 system is stable.**	40.  **In the graph where K > 1 and T > 2 that is the region of stability.**	41. **(i) 0 < K < 1.49 (ii) When K = 1.49 then system is marginally stable. And the frequency of sustained oscillation =** $\frac{1}{\sqrt{10}} rad/sec.$	42. **The range of K for stability: 0 < K < 4**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs